Practice Set 8 Paper 1 – Basic Sciences in Urology

1. Q1. Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria.
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Practice paper compiled from the question bank.

Q1. Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria.

Type: LAQ (25 marks) (also answered in 2018-Summer Paper 1, Q1 and 2024-Winter Paper 3, Q2)

General plan of the lymphatic system

- Lymph flows **cephalad and right to left** and returns to the left innominate (brachiocephalic) vein.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) drain to common iliac nodes, then to **lumbar nodes** and the lumbar trunks. Pelvic lymph crosses freely between the two sides.
- **Lumbar groups:** left lumbar (preaortic, left para-aortic, retroaortic), interaortocaval, and right lumbar (precaval, paracaval, retrocaval). They drain organs supplied by lateral aortic branches: kidneys, adrenals, ureters and gonads.
- Preaortic nodes (coeliac, superior and inferior mesenteric) form the **intestinal trunk**, which carries **chyle** from the intestinal lacteals.
- Lumbar and intestinal trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). It continues as the **thoracic duct**, which rises behind the aorta to the left innominate vein.

Lymphatic drainage of the urinary tract

Organ	Intrinsic lymphatics	Regional nodes
Kidney	Superficial capsular and deep hilar networks along the interlobular, arcuate and interlobar arteries, to nodes at the renal vein	**Left:** left lateral para-aortic (sometimes retrocrural, or thoracic duct above the diaphragm). **Right:** interaortocaval and right paracaval (may cross to left para-aortic)
Ureter	Plexuses in the muscle and adventitia	Abdominal: left para-aortic (left); paracaval and interaortocaval (right). Middle: common iliac. Pelvic: common, external and internal iliac
Bladder	Begins in the lamina propria; a few perivesical nodes	Mainly **external iliac**; also internal iliac and obturator; cross-drainage to both sides
Prostate	Periprostatic plexus	Obturator and internal iliac; presacral; external iliac
Urethra	Submucosal plexus	Male prostatic and membranous: internal iliac, obturator; penile: inguinal, then external iliac. Female: proximal to pelvic, distal to inguinal nodes

- **Applied:** upper tract cancers spread to lumbar nodes; bladder and prostate cancers to pelvic nodes. Renal drainage is asymmetric: right renal lymph can cross left; left stays left.
- Valves in the renal hilar lymphatics normally stop backflow from the cisterna chyli. **Failure of these valves is the basis of chyluria.**

Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons, triglyceride and protein) in urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (**pyelolymphatic fistula**).

Etiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; *Culex* mosquito)	Congenital lymphangiectasia, renal lymphangioma
Brugia (lymphatic filariasis that spares the genitals; rarely urological)	Trauma, surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy

Pathogenesis

- **Life cycle:** mosquito-borne larvae mature into adult worms that live for years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive (reflux) theory:**

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Adult worms + host immune reaction -> lymphangitis and fibrosis
 -> obstruction between intestinal trunks/cisterna chyli and thoracic duct
 -> dilatation of retroperitoneal lymphatics, valve incompetence
 -> reflux of chyle into renal hilar and peripelvic lymphatics
 -> raised pressure ruptures thin-walled lymphatics into a calyceal fornix
 -> pyelolymphatic fistula -> chyluria
```

- The fistula is usually renal (hilar lymphatics lie against thin fornices); rarely ureteric or vesical.
- Chyluria appears **earlier than genital elephantiasis**, is usually **intermittent** and may remit spontaneously. It worsens after fatty meals, when lacteal flow is highest.
- **Non-parasitic causes** work the same way: lymphatic obstruction (tumour, fibrosis), direct disruption (surgery, trauma) or congenital lymphangiectasia.
- **Effects:** loss of fat, protein and lymphocytes causes weight loss, hypoproteinaemia and lowered immunity. Fibrin in chyle forms clots.

Clinical features

- Milky urine after fatty meals; chylous clots (colic, retention); haematochyluria; weight loss, oedema. **Severity:** mild (intermittent), moderate (persistent, clots), severe (haematochyluria, hypoalbuminaemia).

Evaluation

- **Ether test:** milky urine shaken with ether clears (fat dissolves); urine triglycerides positive. Exclude phosphaturia (clears with acetic acid).
- Night blood smear (often negative), filarial antigen test.
- **Cystoscopy after a fatty meal** lateralises the milky efflux; retrograde pyelography or **lymphoscintigraphy** shows the fistula.

Management (outline)

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Mild               -> low-fat, high-protein diet + MCT oil + DEC
Moderate/persistent -> endoscopic sclerotherapy (silver nitrate 0.5-1%, povidone-iodine)
Severe or failed   -> renal hilar lymphatic disconnection (open, laparoscopic, robotic)
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- MCT (medium-chain triglyceride) is absorbed into portal blood, not lacteals, so less chyle forms. DEC (diethylcarbamazine) kills microfilariae.

Diagram to draw: retroperitoneal lymphatics with cisterna chyli, a block above it, and reflux to a pyelolymphatic fistula at a calyceal fornix.

Key points to remember:

- Lymph flows right to left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually *W. bancrofti*.
- Ether test clears milky urine; disconnection surgery for severe or refractory cases.

Q2. Pathophysiology of Renovascular hypertension

Type: LAQ (25 marks) (also answered in 2022-Summer Paper 2, Q2, with treatment in detail)

Definitions

- **RVH (renovascular hypertension):** hypertension caused by RAS (renal artery stenosis) and relieved by revascularisation or nephrectomy. RAS is common in older vascular patients; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS	FMD (fibromuscular dysplasia)
Share	Most (about 90%)	Most of the rest
Patient	Older, smokers, diabetic, vascular disease	Young women
Site	**Ostium and proximal third**	**Mid and distal artery, branches**; "string of beads" (medial fibroplasia)
Course	Progresses to occlusion and renal loss	Rarely occludes

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), aneurysm, dissection, embolism, transplant RAS, **Page kidney** (perinephric compression, e.g. subcapsular haematoma).

The RAAS (renin–angiotensin–aldosterone system)

- **Renin** is released by juxtaglomerular cells of the afferent arteriole in response to:
  1. low afferent arteriolar pressure (intrarenal baroreceptor);
  2. low sodium chloride delivery to the **macula densa**;
  3. sympathetic **β1** stimulation.
- Renin cleaves hepatic **angiotensinogen** to angiotensin I. **ACE (angiotensin-converting enzyme)**, mainly in lung endothelium, converts it to **angiotensin II**.
- **Angiotensin II (AT1 receptor) actions:** systemic arteriolar vasoconstriction; **aldosterone** release (sodium retention, potassium loss); proximal tubular sodium reabsorption; thirst and ADH (antidiuretic hormone); sympathetic facilitation; **efferent arteriolar constriction**, which maintains GFR; cardiac and vascular hypertrophy and fibrosis.
- Normally, rising blood pressure (BP) and sodium suppress renin (negative feedback). In RVH the stenosed kidney "sees" low pressure, so this feedback fails.

Goldblatt experiments

- **Goldblatt (1934)** clipped the canine renal artery and produced sustained hypertension, reversed by early clip removal.

	Two-kidney, one-clip (2K1C)	One-kidney, one-clip (1K1C)
Human equivalent	Unilateral RAS, normal opposite kidney	Bilateral RAS or stenosed solitary kidney
Sodium	Normal kidney excretes extra sodium (**pressure natriuresis**)	No normal kidney: sodium and water retained

	Two-kidney, one-clip (2K1C)	One-kidney, one-clip (1K1C)
Renin	**High** from stenotic side; contralateral suppressed	Rises, then normalises as volume expands
Type	**Renin (angiotensin)-dependent**	**Volume-dependent**
ACE inhibitor effect	BP falls; captopril tests positive	Less BP effect; may cause **acute kidney injury**

Phases of 2K1C hypertension

1. **Acute phase:** renin-dependent; ACE inhibitor or clip removal reverses it.
2. **Transition:** sodium retention and secondary hyperaldosteronism; renin falls towards normal.
3. **Chronic phase:** hypertension damages the contralateral kidney (nephrosclerosis) and resistance vessels. BP no longer depends on the stenosis, so **revascularisation cures less**. This explains poorer results in long-standing hypertension.

Other contributing mechanisms

- Renal afferent nerve signals raise **sympathetic activity**.
- Angiotensin II generates **oxidative stress** (reactive oxygen species), lowers **nitric oxide** and raises **endothelin**, worsening vasoconstriction.
- Inflammation and fibrosis; left ventricular hypertrophy and diastolic dysfunction.

Pathophysiology of ischaemic nephropathy

- The kidney receives about **20–25% of cardiac output**, far more than its metabolic need. **Autoregulation** holds GFR until stenosis is severe (about 70% or more, or a significant translesional pressure gradient).
- Beyond this, GFR depends on angiotensin II-mediated efferent constriction. Blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR: hence **creatinine rise >30%** after these drugs.
- Chronic hypoperfusion causes hypoxia, inflammation, microvascular rarefaction, **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive injury add damage.
- Early disease is reversible ("hibernating" kidney); established fibrosis is not.
- **Flash pulmonary oedema:** bilateral RAS causes volume retention plus angiotensin II-driven diastolic dysfunction.

Clinical clues

- Onset <30 years (FMD) or severe hypertension after 55; **resistant** (≥3 drugs) or accelerated hypertension; sudden loss of control.
- Flash pulmonary oedema; kidneys differing by >1.5 cm; abdominal bruit; **hypokalaemia** (secondary hyperaldosteronism).

Investigations based on pathophysiology

- **Duplex Doppler:** high peak systolic velocity, renal/aortic ratio >3.5. **RI (resistive index) >0.8** means parenchymal disease and poor response.
- **CT (computed tomography) or MR (magnetic resonance) angiography** for anatomy; **catheter angiography** with translesional gradient is the gold standard.
- **Captopril renography:** removing angiotensin II drops GFR in the stenosed kidney (delayed uptake), proving renin-dependence.
- **Renal vein renin ratio >1.5** lateralises.

Treatment implications (brief)

- **ACE inhibitor or ARB** treats renin-dependent RVH (monitor creatinine and potassium), plus statin, antiplatelet and smoking cessation.
- **ASTRAL (2009)** and **CORAL (2014):** stenting added no benefit over medical therapy in atherosclerotic RAS.
- **FMD:** balloon angioplasty without stent. Revascularise atherosclerotic RAS for flash pulmonary oedema, rapidly falling function or truly resistant hypertension.

Diagram to draw: RAAS cascade from renin release to angiotensin II actions, beside the 2K1C and 1K1C models.

> **Key points to remember:**
>
> - Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
> - Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
> - Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
> - ASTRAL and CORAL: no routine stenting; FMD = angioplasty alone.

Q3a. Lymphocele

Type: SAQ (10 marks)

Definition

- A collection of lymph **without an epithelial lining** (a pseudocyst with a fibrous wall), caused by divided lymphatics that keep leaking. It is the **commonest complication of PLND (pelvic lymph node dissection)**.

Settings and incidence

- **PLND** (radical prostatectomy, cystectomy): radiological in up to 60%, clinically significant 2–5%. An extended template roughly doubles the rate.
- **Renal transplantation:** from lymphatics around the recipient iliac vessels or donor hilum. Radiological 0.6–33.9%; most appear within 6 months, peaking at 6 weeks.
- Also after RPLND (retroperitoneal lymph node dissection), inguinal dissection, nephrectomy.
- **Risk factors:** extensive dissection, obesity, heparin prophylaxis, mTOR (mammalian target of rapamycin) inhibitors, acute rejection, retransplantation.

Clinical features

- Most are silent. Large ones compress:
  - iliac vein: leg oedema, **DVT (deep vein thrombosis)**;
  - bladder: frequency, urgency, pelvic pressure;
  - ureter or graft: hydronephrosis, rising creatinine.
- Infection: fever, rigors, sepsis.

Diagnosis

- **Ultrasound:** anechoic, often septated, thin-walled collection. **CT (computed tomography)** maps it before intervention.
- **Aspirate:** lymph has creatinine equal to serum (urinoma: much higher), lymphocytes; culture excludes abscess.

Management

Situation	Treatment
Asymptomatic, <5 cm	Observe; most resolve
Symptomatic	**Percutaneous drain** ± sclerotherapy (povidone-iodine, doxycycline)
Recurrent or thick-walled	**Laparoscopic fenestration (marsupialisation)** into a peritoneal window
Infected	Antibiotics + percutaneous drainage; exclude urine leak
Laparoscopy unsafe	Open drainage

- **Success:** laparoscopic fenestration best, open next, drain or sclerotherapy intermediate, **aspiration alone worst** (reaccumulates, risks infection).
- Fenestration lets the peritoneum absorb lymph (internal drainage): the definitive cure.
- **Lymphangiography with embolisation:** high success; long-term data awaited.

Prevention

- Clip or ligate visible lymphatics (clips often preferred; no robust data). Spare lymphatics over the external iliac artery (leg lymph).
- **Peritoneal interposition flap** fixed to the bladder reduces clinically significant lymphocele. Transperitoneal access is **not** protective (fluid loculates within 48–72 h).

- Drain after transplantation and major nodal dissections.

Diagram to draw: pelvic lymphocele compressing the external iliac vein and bladder, with a laparoscopic peritoneal window.

> **Key points to remember:**
> - Commonest PLND complication; radiological up to 60%, significant 2–5%.
> - Lymph creatinine = serum; urinoma creatinine higher.
> - Laparoscopic fenestration best; aspiration alone worst.

Q3b. Docetaxel

Type: SAQ (10 marks)

Introduction

- Semisynthetic **taxane** cytotoxic. Since **TAX 327 (2004)** it is the standard first-line chemotherapy for mCRPC (metastatic castration-resistant prostate cancer).

Mechanism of action

- Binds **β-tubulin** and **stabilises microtubules** (GTP [guanosine triphosphate]-independent polymerisation). Cells arrest in **G2/M**.
- Induces apoptosis independent of TP53, through **BCL2 phosphorylation** and p27 induction.
- Prostate-specific: blocks microtubule transport of ligand-bound **AR (androgen receptor)** into the nucleus.

Dose and administration

- **75 mg/m² IV (intravenous) every 3 weeks with prednisone 5 mg twice daily**; 6 cycles in mHSPC (metastatic hormone-sensitive prostate cancer). Steroid premedication limits fluid retention and hypersensitivity. Hepatic metabolism: caution in liver dysfunction.

Uses in urology

Setting	Evidence
mCRPC	**TAX 327:** overall survival (OS) 18.9 vs 16.4 months versus mitoxantrone (HR [hazard ratio] 0.76); better pain and PSA (prostate-specific antigen) response. SWOG 9916 (with estramustine) also positive
mHSPC	**CHAARTED:** OS 57.6 vs 44.0 months (HR 0.61), benefit in **high volume**. STAMPEDE: OS HR 0.78. GETUG-AFU 15 not significant
Triplet	ADT (androgen deprivation therapy) + docetaxel + abiraterone (**PEACE-1**) or darolutamide (**ARASENS**) improves OS
High-risk localised	RTOG 0521: added to RT (radiotherapy) + long-term ADT, OS HR 0.69
Bladder	Second-line single agent (response about 13%); **intravesical gemcitabine–docetaxel** for BCG (bacillus Calmette–Guérin)-unresponsive or BCG-naive high-grade NMIBC (non-muscle-invasive bladder cancer)
Penis	**TPF** (docetaxel, cisplatin, 5-fluorouracil)

CHAARTED high volume: ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Toxicity

- **Neutropenia** and febrile neutropenia (dose-limiting), fatigue, **fluid retention** (peripheral oedema), **peripheral neuropathy**, hyperlacrimation, **nail dystrophy**, alopecia, liver dysfunction, hypersensitivity.
- 3-weekly: more neutropenia; weekly: more fatigue, diarrhoea, tearing.

Resistance and sequencing

- P-glycoprotein efflux pump mediates resistance. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel (**TROPIC**; **CARD** beat a second ARPI [androgen receptor pathway inhibitor]).

- **Also know:** failed combinations (bevacizumab, atrasentan).

Diagram to draw: microtubule stabilised by docetaxel causing G2/M arrest and blocking AR entry into the nucleus.

> **Key points to remember:**
> - 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
> - CHAARTED: benefit in high-volume mHSPC (HR 0.61).
> - Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

Q3c. Cephalsporins

Type: SAQ (10 marks)

Assumed meaning: cephalosporins (generations, spectrum, urological use).

Introduction and mechanism

- **β-lactam** antibiotics. They bind **PBPs (penicillin-binding proteins)** and block peptidoglycan cross-linking, so the cell wall fails: **bactericidal**, time-dependent killing.
- Mostly renal excretion (adjust dose in renal failure); **ceftriaxone** also has biliary excretion (no adjustment).
- Resistance: β-lactamases (**ESBL [extended-spectrum β-lactamase]**, AmpC), altered PBPs, reduced porin entry.

Generations

Generation	Examples	Spectrum	Urological use
First	Cefazolin (intravenous), cephalexin, cefadroxil (oral)	Gram-positive (not MRSA [methicillin-resistant *Staphylococcus aureus*]); *E. coli*, *Klebsiella*, *P. mirabilis*	**Surgical prophylaxis (cefazolin)**; cystitis in pregnancy; paediatric prophylaxis
Second	Cefuroxime, cefaclor; cefoxitin, cefotetan	Wider Gram-negative; cefoxitin and cefotetan add **anaerobes**	Prophylaxis for vaginal and bowel surgery; SWL (shock-wave lithotripsy)
Third	Ceftriaxone, cefotaxime, cefixime, cefpodoxime; **ceftazidime**	Most Enterobacterales; **only ceftazidime covers *Pseudomonas***	Pyelonephritis, urosepsis; **gonorrhoea (ceftriaxone 500 mg IM [intramuscular])**; prostate biopsy prophylaxis
Fourth	Cefepime	Gram-negatives incl. *Pseudomonas*, stable to AmpC; Gram-positive	Complicated, nosocomial UTI (urinary tract infection)
Fifth	Ceftaroline	MRSA + Gram-negatives	Limited urological role

- **Newer combinations** for complicated UTI with multidrug resistance: ceftazidime–avibactam, ceftolozane–tazobactam, cefiderocol.

Gaps in spectrum

- **Enterococci** (intrinsically resistant; never treat with cephalosporins), *Listeria*, MRSA (except ceftaroline), atypical organisms.
- ESBL and AmpC producers hydrolyse third-generation agents.

Adverse effects

- Hypersensitivity, less than penicillins. Cross-reactivity with third-generation agents is <1%; avoid after immediate (IgE [immunoglobulin E]-mediated) penicillin reaction.
- ***C. difficile* colitis**; "collateral damage" (selection of ESBL organisms).
- Positive Coombs test; bleeding with cefotetan and cefamandole.
- Ceftriaxone: biliary sludge; avoid in jaundiced neonates and with intravenous calcium.

Paediatric use

- Febrile UTI in neonates: **ampicillin + third-generation cephalosporin** (covers enterococcus). Oral cefixime 8 mg/kg/day. A first-generation agent is a good narrow-spectrum choice for many outpatient UTIs.

Diagram to draw: β-lactam ring binding a PBP and blocking peptidoglycan cross-links.

Key points to remember:

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Q3d. USG in neonates

Type: SAQ (10 marks)

Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up.

Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.
- **Technique:** higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.

Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver; **prominent hypoechoic pyramids mimic dilated calyces or cysts**.
- **Dilation is under-read in the first 48 hours** (physiological dehydration).

Antenatal hydronephrosis follow-up

- **First scan after 48 hours**; earlier if solitary kidney or suspected PUV (posterior urethral valves).
- Normal postnatal **APD (anteroposterior pelvic diameter) <10 mm**, no calyceal or ureteric dilation.

UTD (urinary tract dilation) risk	Findings	Action
P1 (SFU [Society for Fetal Urology] 1–2)	APD 10–15 mm, central calyces	Repeat US at 3–6 months; no VCUG or prophylaxis
P2 (SFU 3)	APD ≥15 mm, peripheral calyces or dilated ureter	Repeat US; VCUG (voiding cystourethrogram) and prophylaxis at discretion
P3 (SFU 4)	Abnormal parenchyma or bladder	Repeat US at 1 month; VCUG and prophylaxis

- **SFU grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.

Other neonatal indications

Condition	Ultrasound finding
PUV	Thick bladder, **keyhole sign**, bilateral hydroureteronephrosis
MCDK (multicystic dysplastic kidney)	Non-communicating cysts of varying size, scant parenchyma
ARPKD (autosomal recessive polycystic kidney disease)	Large, bilaterally echogenic kidneys
Masses	Mesoblastic nephroma; adrenal haemorrhage; renal vein thrombosis (large kidney, hypertension)
Duplex anomalies	Ureterocele, ectopic ureter, megaureter
Spine	Occult dysraphism before arch ossification (up to 6 months)
DSD (differences of sex development)	Müllerian structures, gonads

Also know: hydrometrocolpos, scrotal masses, contrast-enhanced voiding urosonography.

Limitations

- Normal US (ultrasound) does not exclude VUR (vesicoureteral reflux). Function needs MAG3 (mercaptoacetyltriglycine) renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

Diagram to draw: SFU grades 1–4 as kidney outlines.

> **Key points to remember:**
> - First postnatal scan after 48 h; normal APD <10 mm.
> - Echogenic cortex and prominent pyramids are normal in neonates.
> - Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Q3e. Interventional urological treatment in urology

Type: SAQ (10 marks)

Assumed meaning: interventional radiology (image-guided percutaneous and endovascular) treatment in urology.

Principles

- Guidance by ultrasound, fluoroscopy or CT (computed tomography), usually under local anaesthesia. **Seldinger technique:** needle, guidewire, then devices over the wire.
- **Beforehand:** correct coagulopathy, manage anticoagulants, urine culture, antibiotic prophylaxis.

Non-vascular procedures

Procedure	Indications and points
PCN (percutaneous nephrostomy)	Obstructed infected kidney (as effective as retrograde stent), malignant ureteric obstruction, urinary leak or fistula, access for PCNL (percutaneous nephrolithotomy). Puncture a **posterior calyx** through the avascular Brödel plane. Transfusion 0.5–4%
Antegrade stenting	When retrograde stenting fails: malignant obstruction, ileal conduit, transplant ureter
Antegrade pyelography, Whitaker test	Define strictures; pressure–flow test for obstruction
Percutaneous drainage	Renal or perinephric abscess >3 cm, urinoma, lymphocele
Image-guided biopsy	Renal mass biopsy

Vascular (endovascular) procedures

Procedure	Indications
Renal angioembolisation	Grade IV–V renal trauma; **pseudoaneurysm or arteriovenous fistula** after PCNL or partial nephrectomy (delayed bleed); bleeding AML (angiomyolipoma, first line); palliative embolisation of RCC (renal cell carcinoma)
Internal iliac (anterior division) embolisation	Intractable bladder haemorrhage; initial control about 90%. Spare the posterior division (gluteal ischaemia)
PAE (prostate artery embolisation)	LUTS (lower urinary tract symptoms) from BPH (benign prostatic hyperplasia); EAU (European Association of Urology) for men accepting lesser outcomes; AUA (American Urological Association) 2023 may offer
Varicocele embolisation	Coils or liquid sclerosants via femoral or jugular vein
High-flow priapism	Superselective cavernosal artery embolisation; resolution ≥75%; recurrence 30–40%
PTRA (percutaneous transluminal renal angioplasty)	Fibromuscular dysplasia; transplant renal artery stenosis

Ablation

- **Cryoablation or RFA (radiofrequency ablation)**, percutaneous, for cT1a renal masses <3 cm; biopsy first. Higher local recurrence than partial nephrectomy, salvaged by repeat ablation.

Complications

- Contrast nephropathy, access-site haematoma, **non-target embolisation**, **post-embolisation syndrome** (fever, pain, leucocytosis; self-limiting).

Diagram to draw: prone PCN through a posterior calyx along the Brödel line.

Key points to remember:

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula.
- Internal iliac anterior division for bladder bleeding; spare posterior division.

Q3f. Familial malignancy in urology

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q3f and Practice Set 2 Paper 1, Q3b)

Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer. Tumours are **early, bilateral and multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

Hereditary kidney cancer syndromes

Syndrome	Gene (chromosome)	Kidney tumour	Other features
VHL (von Hippel–Lindau)	VHL (3p25)	Clear cell, multifocal	Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma
HPRC	MET (7q31)	Multiple bilateral papillary type 1	None
HLRCC (hereditary leiomyomatosis and RCC)	FH (fumarate hydratase, 1q42–43)	**Aggressive** papillary, collecting duct	Skin and uterine leiomyomas
BHD (Birt–Hogg–Dubé)	FLCN (17p11.2)	Chromophobe, hybrid oncocytic	Fibrofolliculomas, lung cysts, pneumothorax
TSC (tuberous sclerosis complex)	TSC1 (9q34), TSC2 (16p13.3)	Multiple AMLs (angiomyolipomas); RCC 2–3%	Angiofibromas, cortical tubers

- **Also know:** SDH (succinate dehydrogenase)-deficient RCC; BAP1; PTEN (Cowden).

Kidney management

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD), saving function for repeat surgery. **Exception: HLRCC**: excise promptly, wide margins.
- TSC AML: everolimus if >3 cm. **Belzutifan** (HIF-2α [hypoxia-inducible factor 2α] inhibitor) is approved for VHL tumours.

Other urological cancers

- **Prostate:** BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13; father and brother affected: 9-fold risk. Earlier screening; **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, 75-fold with MSH2. Screen with MMR (mismatch repair) immunohistochemistry.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary (MEN-2 [multiple endocrine neoplasia type 2], VHL, NF1 [neurofibromatosis type 1], SDHB).
- **Also know:** Wilms tumour (WT1 in WAGR [Wilms, aniridia, genitourinary anomalies, retardation] and Denys–Drash; Beckwith–Wiedemann).

Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screen relatives.

Diagram to draw: VHL loss → HIF accumulation → VEGF (vascular endothelial growth factor) → angiogenesis.

> **Key points to remember:**
>
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

Rapid Revision Sheet – Practice Set 8 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria. (LAQ)

- Lymph flows right to left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually *W. bancrofti*.
- Ether test clears milky urine; disconnection surgery for severe or refractory cases.

Q2. Pathophysiology of Renovascular hypertension (LAQ)

- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- ASTRAL and CORAL: no routine stenting; FMD = angioplasty alone.

Q3a. Lymphocele (SAQ)

- Commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best; aspiration alone worst.

Q3b. Docetaxel (SAQ)

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).
- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

Q3c. Cephalsporins (SAQ)

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

Q3d. USG in neonates (SAQ)

- First postnatal scan after 48 h; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Q3e. Interventional urological treatment in urology (SAQ)

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula.
- Internal iliac anterior division for bladder bleeding; spare posterior division.

Q3f. Familial malignancy in urology (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.